



Mobile Application System Design

(Chap 3 Part 1)

Key Concepts:

1. Widgets and Widgets Trees
2. Function and Arguments
3. Property and Values
4. MaterialApp and Scaffold
5. Container, Center & Text Widgets
6. const and final
7. Stateless and Stateful Widgets

Widget & Widget Trees

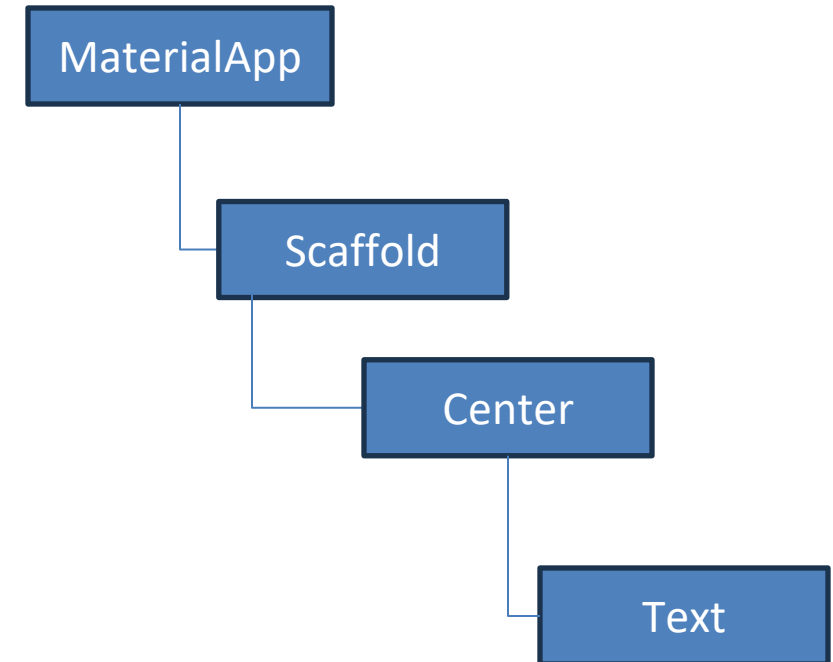
Widget – Small building block of your app

- Visible : things you see (text, image, button)
- Invisible : things that hold other widgets (Container, Column, Center)

Widget Tree – The structure connecting all widgets together

Parent widget: The one that *contains* another widget

Child widget: The one *inside* the parent



Properties and Values

Property – Setting or option for a widget. It tells Flutter what this widget should look like or do

- For example: color

Value – What you assign to a property

- For example: blue

```
WidgetName(  
  propertyName: value,  
)
```

QUESTION ?

“If I want to make a screen with pink background and white text in the center that says ‘Welcome’, what are the widgets, properties, and values I’d use?”

MaterialApp and Scaffold Widgets

MaterialApp – Main container of your whole app

Scaffold – What gives your app its basic screen structure

MaterialApp

└ Scaffold

└ appBar: Title of the app

└ body: Main screen content

└ floatingActionButton: Button for quick actions

└ drawer: Side menu

Container, Center & Text Widgets

Container Widget – A box that you can style, color, resize, or even decorate

Center Widget – Centers whatever is inside it

Text Widget – How we display any text or label on the screen

const and final

const – The value will never change

- For example:

```
const Text ("Welcome")  
const Color (0xFF2196F3)
```

final – The value cannot change but the difference is, the value is decided later, when the app is running

- For example:

```
final DateTime today = DateTime.now();
```

Stateless vs Stateful Widgets

Stateless Widget – Something that never changes once it's built on the screen

- For example:
 - Logo
 - Tittle text
 - Background image
 - Fixed button

Stateful Widget – Something that can change while the app is running

- For example:
 - Login form
 - Play/pause button

CLASS ACTIVITY

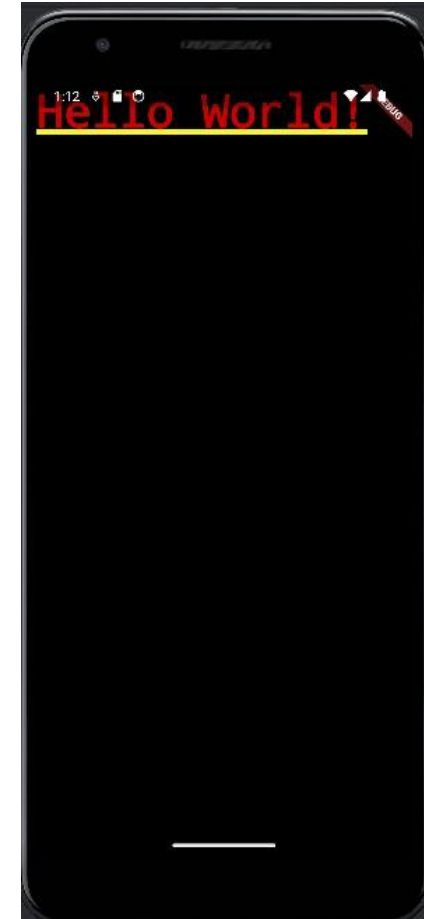
(Hands-On Practice)

Building Your First App

- This week you will start building your first app where a lot of new mobile development concepts will be covered
 - **Widget and Widget Trees (e.g. text, image, button etc)**
 - **Function and Arguments**
 - **const and final value**
 - **stateless and stateful widget**

Example 1 App

```
main.dart x
1  import 'package:flutter/material.dart';
2
3  >> void main(){
4      runApp(
5          const MaterialApp(home: Text('Hello World!')));
6  }
```



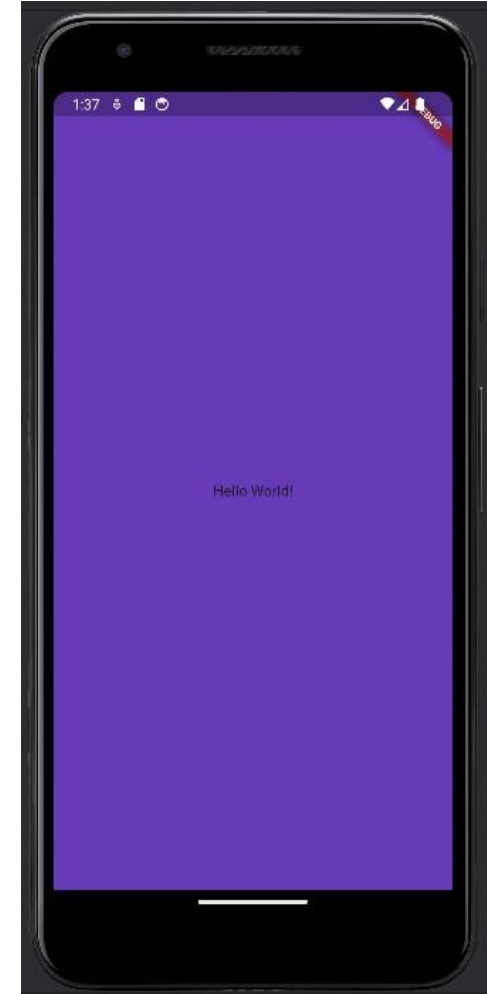
Example 2 App

```
main.dart x
1  import 'package:flutter/material.dart';
2
3  >> void main(){
4      runApp(
5          const MaterialApp(
6              home: Scaffold(
7                  body: Center(
8                      child: Text('Hello World')
9                  ), // Center
10             ), // Scaffold
11         ), // MaterialApp
12     );
13 }
```



Example 2 App

```
main.dart x
1  import 'package:flutter/material.dart';
2
3  >> void main(){
4      runApp(
5          const MaterialApp(
6              home: Scaffold(
7                  backgroundColor: Colors.deepPurple,
8                  body: Center(
9                      child: Text('Hello World!')
10                 ), // Center
11             ), // Scaffold
12         ), // MaterialApp
13     );
14 }
```



Example 2 App

```
main.dart x
1 import 'package:flutter/material.dart';
2
3 void main(){
4   runApp(
5     const MaterialApp(
6       home: Scaffold(
7         backgroundColor: Colors.deepPurple,
8         body: Center(
9           child: Text('Hello World!')
10         ), // Center
11       ), // Scaffold
12     ), // MaterialApp
13   );
14 }
```

```
const MaterialApp(
  home: Scaffold(
    backgroundColor: Colors.|
    body: Center(
      child: Text('Hello
    ), // Center
  ), // Scaffold
), // MaterialApp
}

deepPurple
amber
amberAccent
black
black12
black26
black38
```

Example 3 App

```
1  import 'package:flutter/material.dart';
2
3  >> void main(){
4    runApp(
5      MaterialApp(
6        home: Scaffold(
7          body: Container(
8            decoration: const BoxDecoration(
9              gradient: LinearGradient(
10               colors: [
11                 Color.fromARGB(255, 26, 2, 80),
12                 Color.fromARGB(255, 45, 7, 98)
13               ]
14             ) // LinearGradient
15           ), // BoxDecoration
16          child: const Center(
17            child: Text('Hello World!')
18          ), // Center
19        ) // Container
20      ), // Scaffold
21    ), // MaterialApp
22  );
23 }
```



Example 3 App

```
1 import 'package:flutter/material.dart';
2
3 void main(){
4   runApp(
5     MaterialApp(
6       home: Scaffold(
7         body: Container(
8           decoration: const BoxDecoration(
9             gradient: LinearGradient(
10              colors: [
11                Color.fromARGB(255, 26, 2, 80),
12                Color.fromARGB(255, 45, 7, 98)
13              ],
14              begin: Alignment.topLeft,
15              end: Alignment.bottomRight
16            ) // LinearGradient
17          ), // BoxDecoration
18          child: const Center(
19            child: Text('Hello World!')
20          ), // Center
21        ) // Container
22      ), // Scaffold
23    ), // MaterialApp
```



BASIC RULE

- 📌 Widget – A building block of UI. Visible or Structural
- 📌 Property – A setting or configuration inside a widget.
- 📌 Value – The data or object passed to the property.

MaterialApp widget

```
MaterialApp(  
  home: Scaffold(  
    // ...  
  )  
)
```

home: → Property of MaterialApp

Scaffold(...) → Widget that is the value of home

Scaffold widget

```
home: Scaffold(  
  body: Container(  
    // ...  
  )  
)
```

body: → Property of Scaffold

Container(...) → Widget that is the value of body

Container, Center & Text widget

```
body: Container(  
  decoration: const BoxDecoration(  
    gradient: LinearGradient(  
      colors: [  
        Color.fromARGB(255, 26, 2, 80),  
        Color.fromARGB(255, 45, 7, 98),  
      ],  
      begin: Alignment.topLeft,  
      end: Alignment.bottomRight,  
    ), // LinearGradient  
  ), // BoxDecoration  
  child: const Center(  
    child: Text(  
      'Hello World!',  
      style: TextStyle(  
        color: Colors.white,  
        fontSize: 28,
```

1. *decoration: (Property of Container)*

- decoration: is the property name
- BoxDecoration(...) is the value or object we pass to apply styling (like gradients, borders)

Inside BoxDecoration

- gradient: is the property
- LinearGradient(...) is the object that creates the color transition

Inside LinearGradient

- colors: → Property of LinearGradient
- Color.fromARGB(...) → Values (they define which colors to blend)

2. *child: (Another property of Container)*

- child: is a property of the Container
- Center(...) is a widget that centers its child

Inside Center: child: Text(...)

- child: is a property of the Center widget
- Text(...) is a widget to display a line of text

Inside Text: Styling with TextStyle

- style: → Property of Text
- TextStyle(...) → Object that controls how the text looks

Example 3 App

```
    COLOR.fromARGB(255, 43, 7, 98)  
  ],  
  begin: Alignment.topLeft,  
  end: Alignment.bottomRight  
) // LinearGradient  
, // BoxDecoration  
child: const Center(  
  child: Text('Hello World!', style: TextStyle(  
    color: Colors.white,  
    fontSize: 28  
  )) // TextStyle, Text  
, // Center  
) // Container  
, // Scaffold  
) // MaterialApp
```



Example 4 App

```
1 import 'package:flutter/material.dart';
2
3 void main(){
4   runApp(
5     const MaterialApp(
6       home: Scaffold(
7         body: GradientContainer()
8       ), // Scaffold
9     ), // MaterialApp
10  );
11 }
12
13 class GradientContainer extends StatelessWidget{
14   const GradientContainer({super.key});
15   @override
16   Widget build(context){
17
18   }
19 }
```

```
class GradientContainer extends StatelessWidget{
  const GradientContainer({super.key});
  @override
  Widget build(context){
    return Container(
      decoration: const BoxDecoration(
        gradient: LinearGradient(
          colors: [
            Color.fromARGB(255, 26, 2, 80),
            Color.fromARGB(255, 45, 7, 98)
          ],
          begin: Alignment.topLeft,
          end: Alignment.bottomRight
        ) // LinearGradient
      ), // BoxDecoration
      child: const Center(
        child: Text('Hello World!', style: TextStyle(
          color: Colors.white,
          fontSize: 28
        )) // TextStyle, Text
      ), // Center
    ); // Container
  }
}
```



Example 4 App

```
main.dart  gradient_container.dart x
1  import 'package:flutter/material.dart';
2
3  class GradientContainer extends StatelessWidget{
4    const GradientContainer({super.key});
5    @override
6    Widget build(context){
7      return Container(
8        decoration: const BoxDecoration(
9          gradient: LinearGradient(
10             colors: [
11               Color.fromARGB(255, 26, 2, 80),
12               Color.fromARGB(255, 45, 7, 98)
13             ],
14             begin: Alignment.topLeft,
15             end: Alignment.bottomRight
16           ) // LinearGradient
17        ), // BoxDecoration
18        child: const Center(
19          child: Text('Hello World!', style: TextStyle(
20             color: Colors.white,
21             fontSize: 28
22           )) // TextStyle, Text
23        ), // Center
24      ); // Container
25    }
26  }
```

gradient_container.dart

```
main.dart x  gradient_container.dart x
1  import 'package:flutter/material.dart';
2  import 'package:week_1_example_4/gradient_container.dart';
3
4  void main(){
5    runApp(
6      const MaterialApp(
7        home: Scaffold(
8          body: GradientContainer()
9        ), // Scaffold
10      ), // MaterialApp
11    );
12  }
13
```

main.dart



Example 4 App

```
main.dart  gradient_container.dart  styled_text.dart x
1 import 'package:flutter/material.dart';
2
3 class StyledText extends StatelessWidget{
4   const StyledText({super.key});
5   @override
6   Widget build(context){
7     return const Text('Hello World!', style:TextStyle(
8       color: Colors.white,
9       fontSize: 28
10    ) // TextStyle
11  ); // Text
12 }
13 }
```

styled_text.dart

```
main.dart  gradient_container.dart x  styled_text.dart
1 import 'package:flutter/material.dart';
2 import 'package:week_1_example_4/styled_text.dart';
3
4 class GradientContainer extends StatelessWidget{
5   const GradientContainer({super.key});
6   @override
7   Widget build(context){
8     return Container(
9       decoration: const BoxDecoration(
10         gradient: LinearGradient(
11           colors: [
12             Color.fromARGB(255, 26, 2, 80),
13             Color.fromARGB(255, 45, 7, 98)
14           ],
15           begin: Alignment.topLeft,
16           end: Alignment.bottomRight
17         ) // LinearGradient
18       ), // BoxDecoration
19       child: const Center(
20         child: StyledText()
21       ), // Center
22     ); // Container
23   }
24 }
```

gradient_container.dart



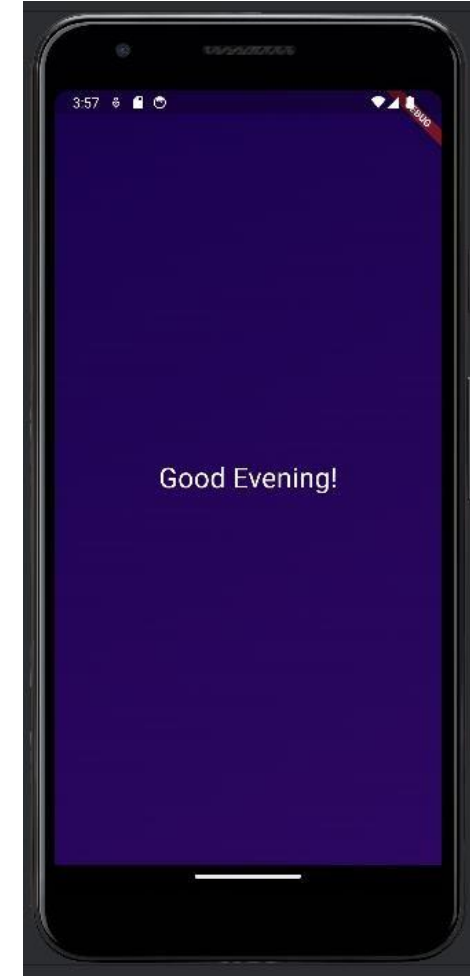
Example 4 App

```
main.dart  gradient_container.dart  styled_text.dart
lib > styled_text.dart > ...
1  import 'package:flutter/material.dart';
2
3  class StyledText extends StatelessWidget{
4
5      final String outputText;
6
7      const StyledText(this.outputText, {super.key});
8
9      @override
10     Widget build(context){
11         return Text(
12             outputText,
13             style: const TextStyle(
14                 color: Colors.white,
15                 fontSize: 28,
16             ), // TextStyle
17         ); // Text
18     }
19 }
```

styled_text.dart

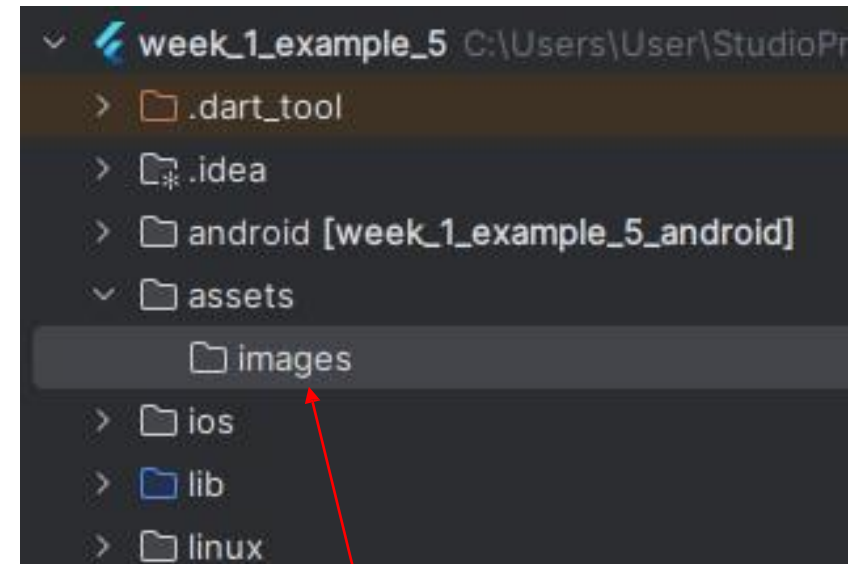
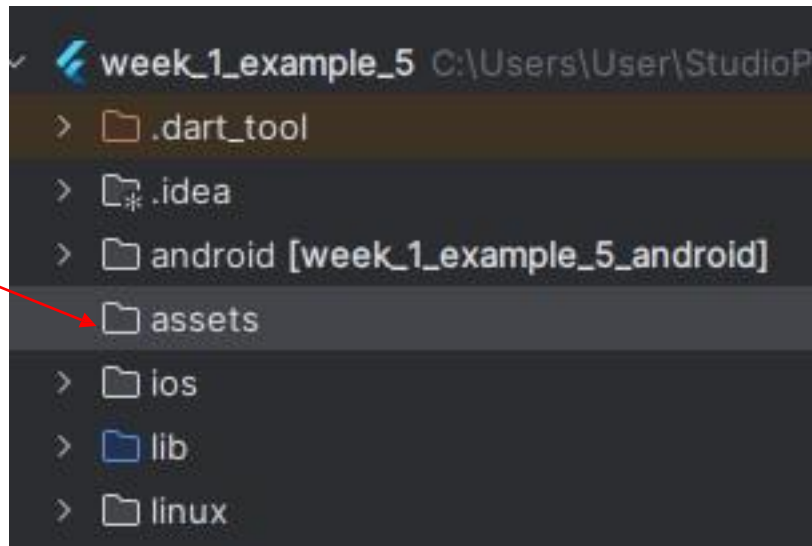
```
main.dart  gradient_container.dart  styled_text.dart
lib > gradient_container.dart > ...
1  import 'package:flutter/material.dart';
2  import 'package:hello_world/styled_text.dart';
3
4  class GradientContainer extends StatelessWidget {
5      const GradientContainer({super.key});
6
7      @override
8      Widget build(BuildContext context) {
9          return Container(
10              decoration: const BoxDecoration(
11                  gradient: LinearGradient(
12                      colors: [
13                          Color.fromARGB(255, 26, 2, 80),
14                          Color.fromARGB(255, 45, 7, 98),
15                      ],
16                      begin: Alignment.topLeft,
17                      end: Alignment.bottomRight,
18                  ), // LinearGradient
19              ), // BoxDecoration
20              child: const Center(
21                  child: StyledText("Good Evening")
22              ), // Center
23          ); // Container
24      }
25 }
```

gradient_container.dart



Example 5 App

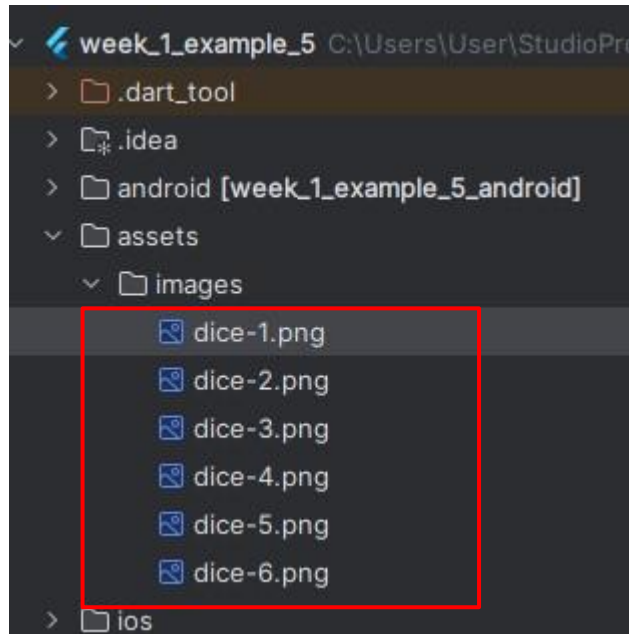
Create a new folder



Create a subfolder

Example 5 App

Remove the “#”



Copy and paste the images to “images” subfolder

```
59 uses-material-design: true
60
61 # To add assets to your application, add an assets section, like this:
62 # assets:
63 # - images/a_dot_burr.jpeg
64 # - images/a_dot_ham.jpeg
65
66 # An image asset can refer to one or more resolution-specific "variants", see
67 # https://flutter.dev/assets-and-images/#resolution-aware
```



```
uses-material-design: true

# To add assets to your application, add an assets section, like this:
assets:
  - images/a_dot_burr.jpeg
  - images/a_dot_ham.jpeg

# An image asset can refer to one or more resolution-specific "variants", see
# https://flutter.dev/assets-and-images/#resolution-aware
```

pubspec.yaml

Example 5 App

```
# To add assets to your application, add an assets section to your pubspec.yaml file.  
assets:  
  - assets/images/dice-1.png  
  - assets/images/dice-2.png  
  - assets/images/dice-3.png  
  - assets/images/dice-4.png  
  - assets/images/dice-5.png  
  - assets/images/dice-6.png  
  
# An image asset can refer to one or more resolution-specific assets.  
# https://flutter.dev/assets-and-images/#resolution-aware
```

pubspec.yaml

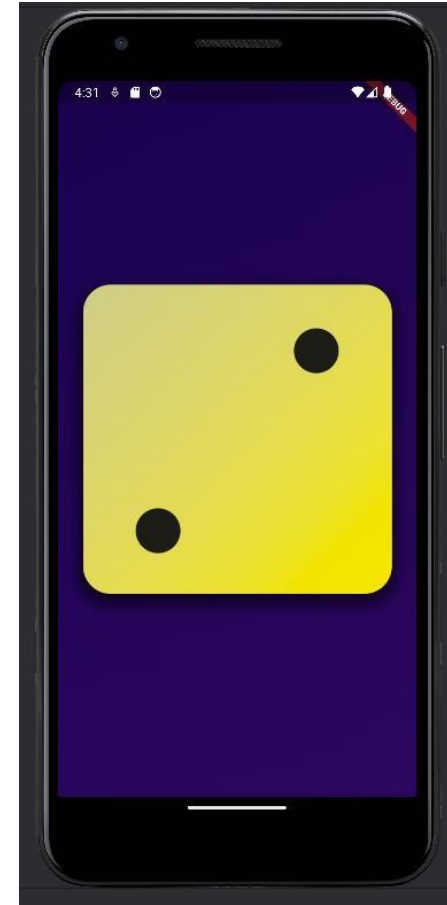
```
main.dart  gradient_container.dart  pubspec.yaml  
1  import 'package:flutter/material.dart';  
2  
3  class GradientContainer extends StatelessWidget {  
4    const GradientContainer({super.key});  
5    @override  
6    Widget build(BuildContext) {  
7      return Container(  
8        decoration: const BoxDecoration(  
9          gradient: LinearGradient(  
10             colors: [  
11               Color.fromARGB(255, 26, 2, 80),  
12               Color.fromARGB(255, 45, 7, 98)  
13             ],  
14             begin: Alignment.topLeft,  
15             end: Alignment.bottomRight  
16           ) // LinearGradient  
17         ), // BoxDecoration  
18         child: Center(  
19           child: Image.asset('assets/images/dice-2.png')  
20         ), // Center  
21       ); // Container  
22     }  
23   }
```

gradient_container.dart

Example 5 App

```
main.dart x gradient_container.dart pubspec.yaml
1 import 'package:flutter/material.dart';
2 import 'package:week_1_example_5/gradient_container.dart';
3
4 void main(){
5   runApp(
6     const MaterialApp(
7       home: Scaffold(
8         body: GradientContainer()
9       ), // Scaffold
10     ), // MaterialApp
11   );
12 }
```

main.dart



Example 5 App

```
main.dart  gradient_container.dart x pubspec.yaml
1  import 'package:flutter/material.dart';
2
3  class GradientContainer extends StatelessWidget{
4    const GradientContainer({super.key});
5    @override
6    Widget build(context){
7      return Container(
8        decoration: const BoxDecoration(
9          gradient: LinearGradient(
10             colors: [
11               Color.fromARGB(255, 26, 2, 80),
12               Color.fromARGB(255, 45, 7, 98)
13             ],
14             begin: Alignment.topLeft,
15             end: Alignment.bottomRight
16           ) // LinearGradient
17        ), // BoxDecoration
18        child: Center(
19          child: Image.asset(
20            'assets/images/dice-2.png',
21            width: 200
22          ), // Image.asset
23        ), // Center
24      ); // Container
25    }
26  }
```

gradient_container.dart



Example 5 App

```
main.dart  gradient_container.dart  pubspec.yaml

1 import 'package:flutter/material.dart';
2
3 class GradientContainer extends StatelessWidget{
4   const GradientContainer({super.key});
5   @override
6   Widget build(context){
7     return Container(
8       decoration: const BoxDecoration(
9         gradient: LinearGradient(
10           colors: [
11             Color.fromARGB(255, 26, 2, 80),
12             Color.fromARGB(255, 45, 7, 98)
13           ],
14           begin: Alignment.topLeft,
15           end: Alignment.bottomRight
16         ) // LinearGradient
17       ), // BoxDecoration
18       child: Center(
19         child: Column(
20           children: [
21             Image.asset(
22               'assets/images/dice-2.png',
23               width: 200
24             ), // Image.asset
25           ]
26         ) // Column
27       ), // Center
28     ); // Container
29   }
30 }
```

gradient_container.dart

```
1 import 'package:flutter/material.dart';
2
3 class GradientContainer extends StatelessWidget{
4   const GradientContainer({super.key});
5
6   void rollDice(){
7
8   }
9
10  @override
11  Widget build(context){
12    return Container(
13      decoration: const BoxDecoration(
14        gradient: LinearGradient(
15          colors: [
16            Color.fromARGB(255, 26, 2, 80),
17            Color.fromARGB(255, 45, 7, 98)
18          ],
19          begin: Alignment.topLeft,
20          end: Alignment.bottomRight
21        ) // LinearGradient
22      ), // BoxDecoration
23      child: Center(
24        child: Column(
25          children: [
26            Image.asset(
27              'assets/images/dice-2.png',
28              width: 200
29            ), // Image.asset
30            TextButton(
31              onPressed: rollDice,
32              child: const Text('Roll Dice')
33            ) // TextButton
34          ]
35        ) // Column
36      )
37    );
38  }
```

gradient_container.dart



Example 5 App

```
@override
Widget build(context){
  return Container(
    decoration: const BoxDecoration(
      gradient: LinearGradient(
        colors: [
          Color.fromARGB(255, 26, 2, 80),
          Color.fromARGB(255, 45, 7, 98)
        ],
        begin: Alignment.topLeft,
        end: Alignment.bottomRight
      ) // LinearGradient
    ), // BoxDecoration
    child: Center(
      child: Column(
        mainAxisSize: MainAxisSize.min,
        children: [
          Image.asset(
            'assets/images/dice-2.png',
            width: 200
          ), // Image.asset
          TextButton(
            onPressed: rollDice,
            child: const Text('Roll Dice')
          ) // TextButton
        ]
      )
    )
  );
}
```

gradient_container.dart



Example 5 App

```
), // BoxDecoration
child: Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.min,
    children: [
      Image.asset(
        'assets/images/dice-2.png',
        width: 200
      ), // Image.asset
      TextButton(
        onPressed: rollDice,
        style: TextButton.styleFrom(
          padding: const EdgeInsets.only(top:20),
          foregroundColor: Colors.white,
          textStyle: const TextStyle(
            fontSize:26
          ) // TextStyle
        ),
        child: const Text('Roll Dice')
      ) // TextButton
    ]
  ) // Column
```

gradient_container.dart



Example 5 App

```
main.dart  gradient_container.dart  dice_roller.dart x
1  import 'package:flutter/material.dart';
2
3  class DiceRoller extends StatefulWidget{
4    const DiceRoller({super.key});
5
6    @override
7    State<DiceRoller> createState(){
8      return _DiceRollerState();
9    }
10 }
11
12 class _DiceRollerState extends State<DiceRoller>{
13   var activeDiceImage = 'assets/images/dice-2.png';
14
15   void rollDice(){
16     setState(){
17       activeDiceImage = 'assets/images/dice-4.png';
18     };
19   }
20
21   @override
22   Widget build(context){
23
24   }
25 }
```

dice_roller.dart

```
void rollDice(){
  setState(){
    activeDiceImage = 'assets/images/dice-4.png';
  };
}

@override
Widget build(context){
  return Column(
    mainAxisAlignment: MainAxisAlignment.min,
    children: [
      Image.asset(
        activeDiceImage,
        width: 200
      ), // Image.asset
      TextButton(
        onPressed: rollDice,
        style: TextButton.styleFrom(
          padding: const EdgeInsets.only(top:20),
          foregroundColor: Colors.white,
          textStyle: const TextStyle(
            fontSize:26
          ) // TextStyle
        ),
        child: const Text('Roll Dice')
      ) // TextButton
    ]
  ); // Column
}
```

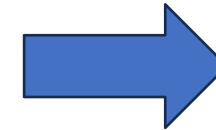
Change to
variable name

dice_roller.dart

Example 5 App

```
main.dart  gradient_container.dart  dice_roller.dart
1  import 'package:flutter/material.dart';
2
3  import 'package:week_1_example_5/dice_roller.dart';
4
5  class GradientContainer extends StatelessWidget{
6    const GradientContainer({super.key});
7
8    @override
9    Widget build(context){
10      return Container(
11        decoration: const BoxDecoration(
12          gradient: LinearGradient(
13            colors: [
14              Color.fromARGB(255, 26, 2, 80),
15              Color.fromARGB(255, 45, 7, 98)
16            ],
17            begin: Alignment.topLeft,
18            end: Alignment.bottomRight
19          ) // LinearGradient
20        ), // BoxDecoration
21        child: const Center(
22          child: DiceRoller()
23        ), // Center
24      ); // Container
25    }
26  }
```

gradient_container.dart



Click



Example 5 App

```
class _DiceRollerState extends State<DiceRoller>{  
  var activeDiceImage = 'assets/images/dice-2.png';  
    
  void rollDice(){  
    var diceRoll = Random().nextInt(6) + 1;  
      
    setState(){  
      activeDiceImage = 'assets/images/dice-$diceRoll.png';  
    });  
  }  
}
```

dice_roller.dart



See you next class